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GrAuFlow: A snakemake workflow for pangenome graph augmentation using short-read data

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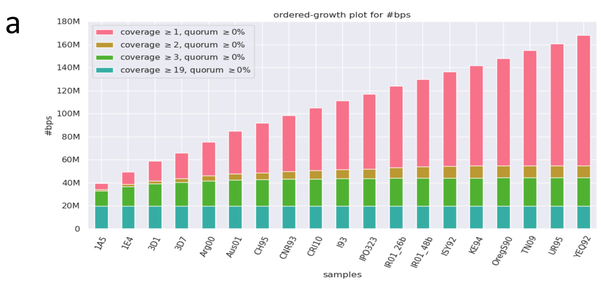
Pangenome graphs and graph augmentation

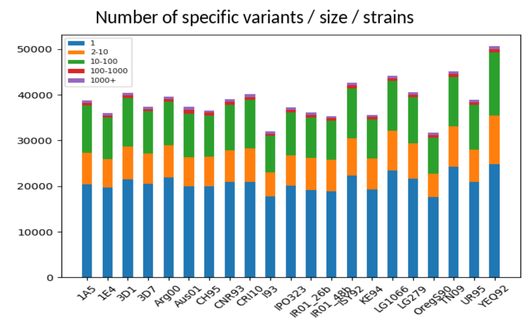
Pangenome graphs (PGGs) are useful tools for representing all genomic variations within a species simultaneously. However, with the constant acquisition of new sequencing data, it is essential to update these graphs to incorporate new genomic resources. The process of adding new genomic data to an existing pangenome graph (PGG) is called graph augmentation. But why use augmentation instead of simply reconstructing the graph from scratch? Tools such as Minigraph can be used to construct a simple pangenome graph including only large variants, but haplotype information is lost in the resulting rGFA. While Minigraph-Cactus could retain such information, the graph would quickly become cluttered and computationally heavy. The augmentation process is used here as a middle ground between these two approaches, with the aim of preserving haplotype information from Minigraph-Cactus while also maintaining the graph structure as simple as possible. By developing GrAuFlow, our Snakemake graph augmentation workflow, we aimed to address the following questions:

- **Can graph augmentation be relevant when dealing with short-read resources for an organism ?**
- **Is it relevant to incorporate all the genetic variability ?**

a



b

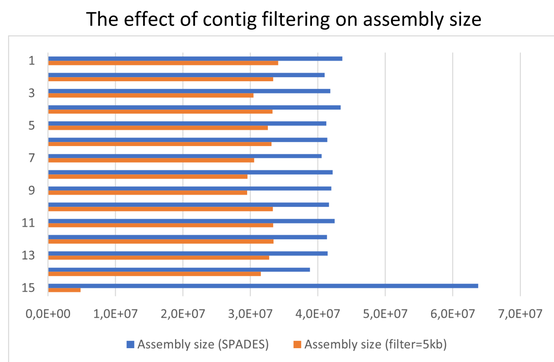


Based on 19 genomes (Badet et al., 2020) and 2 additional unpublished long-read genomes (LG279, LG1066) from the CASDAR SEPTODUR project, we constructed a pangenome graph (PGG) of the wheat fungal pathogen *Zymoseptoria tritici* using Minigraph-Cactus. This graph highlights a core genome of 20 Mb out of an average genome length of 40 Mb. Each genome added to the graph contributes nearly 10 Mb of specific bases (a), but the majority of the added nodes are SNPs or small InDels (b).

To evaluate the approach of graph augmentation, we tested GrAuFlow with a first small set of 15 short-read datasets obtained from individual sampling from the same population (Amezrou et al., 2024).

Short read assembly with SPAdes

The first step is genome assembly using SPAdes to obtain longer sequences. We then developed a Python script to filter out contigs based on criteria such as length, coverage, and GC content to remove contigs with possible assembly errors. For most genomes, we obtain assemblies close to 30 Mb with sequences longer than 5 kb. The long contigs are then split into sequences of 15 kb and mapped onto the graph with Giraffe-lr.



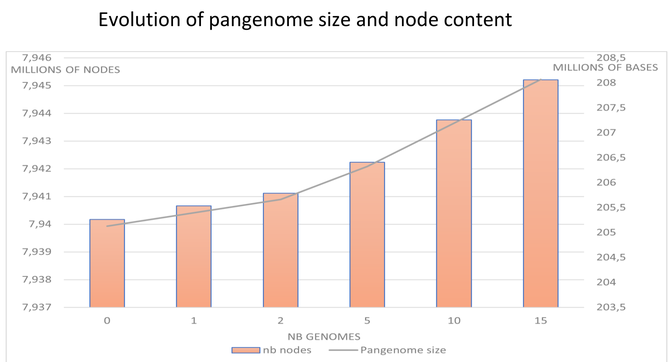
Filtering GAF files

```
cs:Z::622*CA:597*CT:415+ATTTGCGAGTAGTGGAT:3606+AC:1379*AG:6107
       622+1+597 +1+415   ins > 10bp      3606+0 +1379+1+6107
                                                   ins < 10bp
            cs:Z::1636+ATTTGCGAGTAGTGGAT:11093
```

Since vg augment does not provide options to limit or select the variants to include in the graph, we developed a simple filter that rewrites GAF files (CS tag and other info) based on a variant size threshold.

Impact of graph augmentation

a



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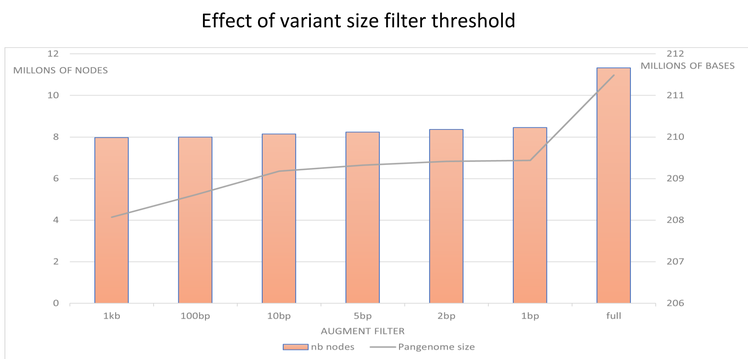


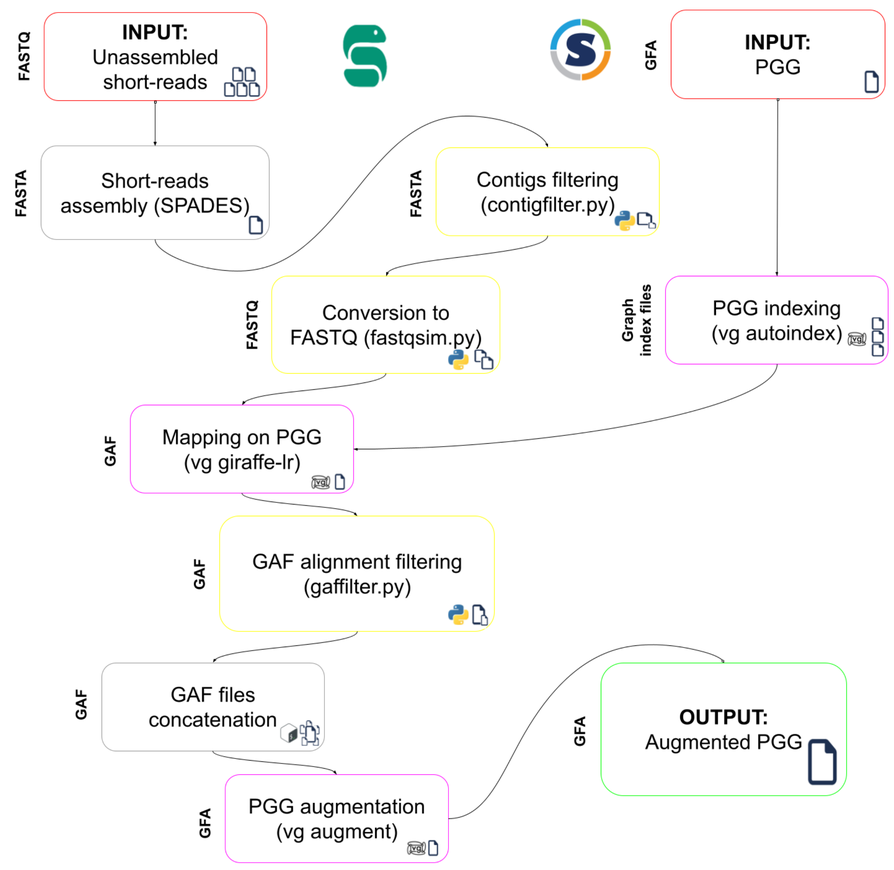
To quantify the impact of graph augmentation, we measured the evolution of the number of nodes and the pangenome size based on the number of genomes added and the size of variants included in the graph. As expected, the size of the pangenome increases with the number of genomes added (a), and it can be greatly reduced simply by removing SNPs (b). Pangenome size growth through augmentation is greatly reduced compared to a whole graph reconstruction approach, but it does not include new haplotypes.

Mapping on augmented graphs with different variant sizes

To evaluate the impact of graph augmentation on graph mapping, we re-mapped the raw reads of one of the samples against different filtering conditions. We observed that the optimal mapping is achieved when all variants are present (no filter), but filtering out small variants can significantly reduce graph complexity while maintaining relatively high mapping performance. Adding InDels only allows for a strong increase in perfect mapping. Nevertheless, whatever the size of the filter, the number of new reads mapped onto the graph remains quite low.

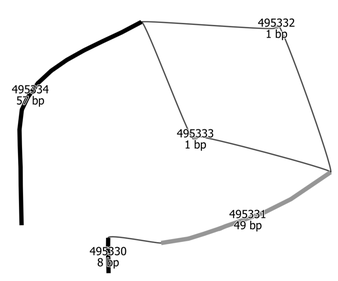
			Graph augmented - variant size filter				
		raw full graph	1kb	100bp	10bp	1bp	no-filter
Total aligned reads		25985896	25999588	26008712	26016003	26021158	26059686
Nb extra-reads mapped onto the graph		0	13692	22816	30107	35262	73790
Total perfect reads		17436551	17674570	17773294	17942854	18099449	19600485
Nb extra-reads perfectly mapped onto the graph		0	238019	336743	506303	662898	2163934
Total properly paired reads		25884256	25902188	25916052	25927008	25935164	25976414
Insertions	nb reads	136273	133201	132014	132837	74711	71127
	nb bases	339579	332706	330793	322491	161069	151447
Deletions	nb reads	222251	218894	216442	169415	93233	88336
	nb bases	628895	619669	613534	380128	210177	190458
Substitutions	nb reads	20443648	19752204	19545672	19234567	19091912	15710533
	nb bases	20479558	19787534	19580678	19268899	19125742	15737421

The different steps of GrAuFlow (Graph Augmentation Workflow) design with Snakemake:



Graph augmentation redefines existing nodes and add new ones

a



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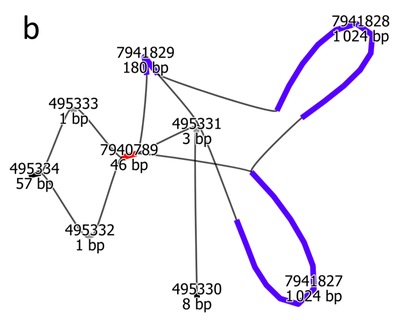

Bandage visualization for a subregion of the pangenome graph before (a) and after (b) augmentation with GrAuFlow. The start and end nodes are colored **black**, the original nodes **grey** and the added nodes **blue**. During the augmentation process, node 495331 (49 bp) was split into nodes 495331 (3 bp) and node 7940789 (46 bp), colored **red** (b). Graph augmentation with vg breaks existing nodes to add new ones when necessary. Existing nodes cannot be fully deleted, in order to preserve the overall graph structure.

Perspectives

Pangenome graph augmentation is an interesting method for achieving pangenome graph scalability by integrating new sequences into an existing graph. With a limited number of genomes, the gain in terms of read mappability remains limited. We now plan to test the augmentation of a graph with over a thousand samples available in the databases, collected from divergent populations of *Z.tritici*, which will provide greater genetic variability. Throughout this work, we have chosen to use the vg giraffe long-read mapper for technical reasons, such as the ability to map sequences on long InDels, which is not possible with tool such as GraphAligner. We also intend to use other long-read mappers, such as PALSS, a tool designed to augment a graph from unassembled long reads. In addition, we will implement strict new criteria for filtering assembled contigs to avoid incorporating chimeric or misassembled regions due to their high repeat content.

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